WEEK 4: CAPSTONE PROJECT REPORT

4-Cylinder Engine Mechanism Assembly and Motion Study Using SOLIDWORKS

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Internship: Mechanical Design Intern – YuvaIntern

Software Used: SOLIDWORKS

1. Introduction

This capstone project focuses on the 3D modelling, assembly, and motion study of a 4-cylinder engine mechanism using SOLIDWORKS. The project demonstrates how individual mechanical components work together to convert the rotary motion of a crankshaft into the reciprocating motion of pistons.

The model includes pistons, piston pins, connecting rods, connecting rod caps, and a crankshaft. The assembly was created to understand component positioning, mechanical mates, and the motion of an engine mechanism.

2. Objectives

- To create and assemble the main components of a 4-cylinder engine mechanism.
- To understand the relationship between the crankshaft, connecting rods, and pistons.
- To apply SOLIDWORKS part modelling and assembly techniques.
- To study the conversion of rotary motion into reciprocating motion.
- To improve practical knowledge of mechanical design and motion studies.

3. Software and Tools Used

- SOLIDWORKS for 3D part modelling and assembly.
- Sketch tools for creating basic component profiles.
- Extruded Boss/Base and Extruded Cut for material addition and removal.
- Revolved Boss/Base for creating suitable rotational features.
- Fillet for improving selected edges and transitions.
- Assembly mates and Motion Study tools for demonstrating the mechanism's movement.

4. Components Modelled

4.1 Crankshaft

The crankshaft is the main rotating component of the mechanism. Its offset crank sections provide the motion needed to move the connecting rods and pistons.

4.2 Pistons

The pistons are the reciprocating components of the mechanism. They move back and forth as the crankshaft rotates.

4.3 Piston Pins

Piston pins connect the pistons to the connecting rods and allow the required relative movement between the components.

4.4 Connecting Rods

Connecting rods link the pistons to the crankshaft. They transmit motion and force between these components.

4.5 Connecting Rod Caps

The connecting rod caps form separate parts of the connecting rod assemblies around the crankshaft journals in the model.

5. Design and Assembly Methodology

1. Created individual 3D component models using sketches and suitable modelling features.
2. Used appropriate features to form the required shapes of the pistons, pins, connecting rods, caps, and crankshaft.
3. Opened the assembly environment and inserted the required components.
4. Positioned the components using suitable assembly mates.
5. Checked the relative positions of the pistons, connecting rods, and crankshaft.
6. Created a motion study to demonstrate the crankshaft rotation and corresponding piston movement.
7. Reviewed the assembly visually to identify positioning or movement issues.

6. Working Principle

The mechanism demonstrates the relationship between the rotary motion of the crankshaft and the reciprocating motion of the pistons.

When the crankshaft rotates, its offset crank sections move the connecting rods. The connecting rods then guide the pistons through their reciprocating movement. The positions of the crank throws determine the relative movement of the four pistons.

The motion study helps visualise the mechanical relationship between the rotating and reciprocating components.

Project Scope: This project focuses on the 3D modelling, assembly, and motion study of the main components of a 4-cylinder engine mechanism using SOLIDWORKS.

7. Final Design and Results

The 3D assembly of the 4-cylinder engine mechanism was created in SOLIDWORKS. The motion study demonstrated the relationship between crankshaft rotation and piston movement.

The project provided practical experience in component modelling, assembly positioning, and visualising mechanical motion.

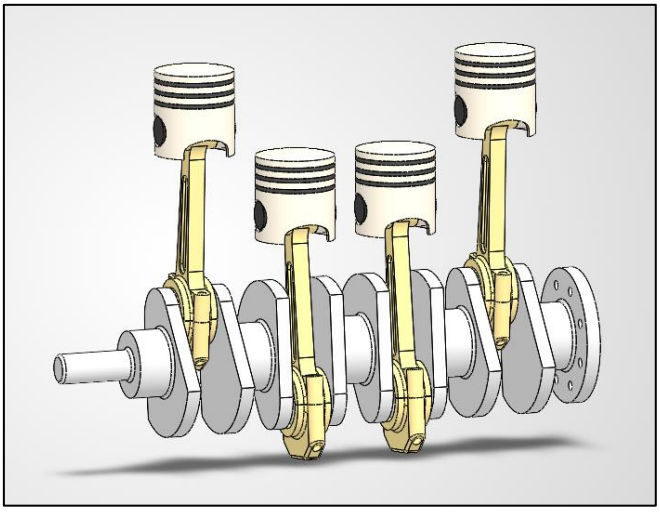


Figure 1: 4-cylinder engine assembly.

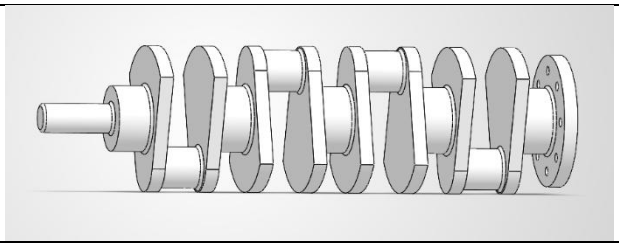


Figure 2: Crankshaft

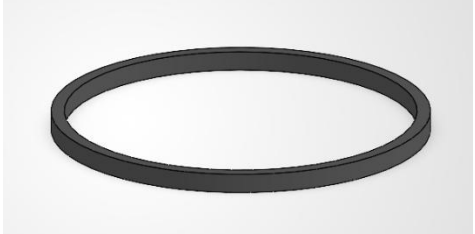


Figure 3: Piston Ring

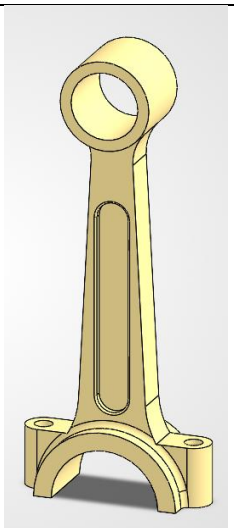


Figure 4: Connecting Rod

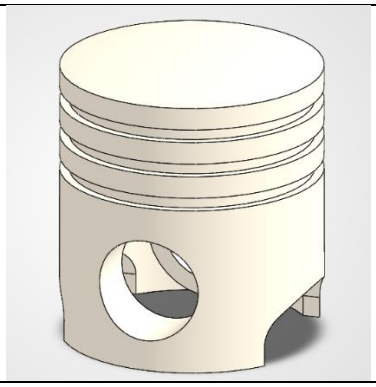
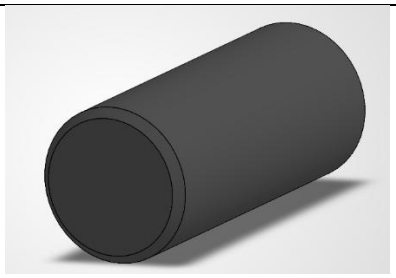
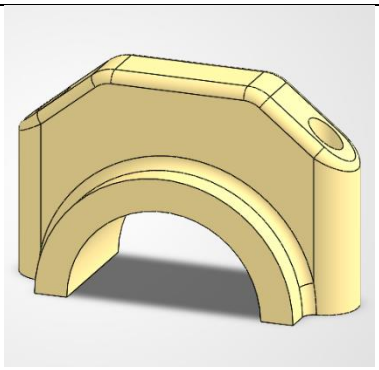
		
Figure 5: Piston	Figure 6: Piston Pin	Figure 7: Connecting Rod Cap

Table 1: Individual components

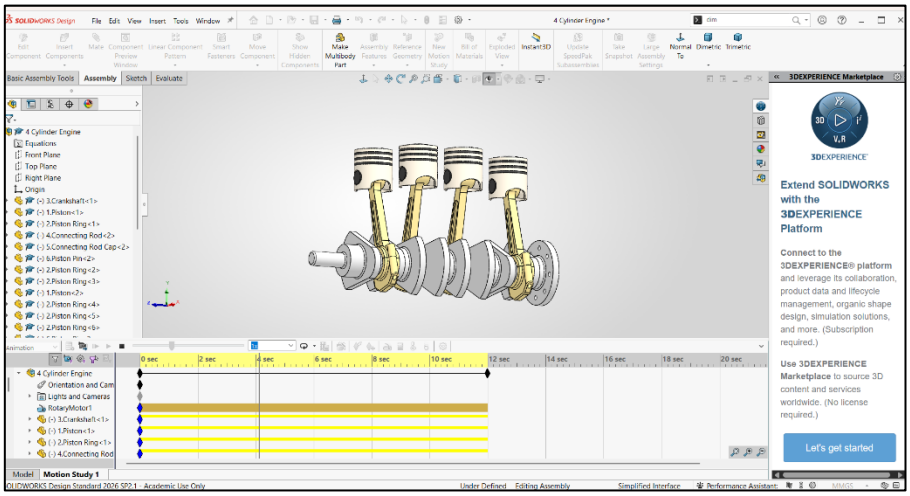


Figure 8: Motion Study screenshot

8. Challenges Faced

During the modelling and assembly process, attention was required to maintain correct component positioning and establish suitable relationships between the moving parts. Understanding how the connecting rods interact with the crankshaft and pistons was also important when setting up the motion study.

These activities helped improve my understanding of mechanical assemblies and the importance of checking component alignment and movement.

9. Learning Outcomes

- Improved knowledge of 3D part modelling using SOLIDWORKS.
- Practical experience in assembling multiple mechanical components.
- Better understanding of assembly mates and component positioning.
- Understanding of rotary-to-reciprocating motion conversion.
- Experience in creating and reviewing a mechanical motion study.
- Improved technical documentation and engineering presentation skills.

10. Future Improvements

The project can be improved by adding a cylinder block, cylinder head, suitable fasteners, and other supporting components. Detailed engineering drawings with dimensions and tolerances can also be prepared. Further improvements could include a more complete assembly motion study and engineering analysis where suitable software and validated design data are available.

11. Conclusion

The 4-cylinder engine mechanism assembly project provided practical experience in mechanical design using SOLIDWORKS. The modelling, assembly, and motion study helped demonstrate the interaction between the crankshaft, connecting rods, and pistons.

This capstone project strengthened my understanding of 3D modelling, mechanical assembly, and motion simulation. It also provided a foundation for further learning in engine mechanisms, detailed engineering drawings, and mechanical design analysis.

Submitted to: YuvaIntern – Mechanical Design Internship